

What else this paper could prove

New theorems worth adding, and what already follows from what is there

Companion to the problem report. All claims worked out against `dhkm.tex`'s own definitions.
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How to read this

Fourteen suggestions, in the order I would act on them. Each one gives the statement, a proof or a proof idea, and one paragraph on why it is worth having. Tags: **[VERIFIED]** means I checked the proof; **[PROOF SKETCH]** means the argument is outlined and rests on standard facts; **[OPEN]** means I am proposing it as a question, not claiming it. **[ALSO FIXES A DEFECT]** marks the ones that also repair a defect from the problem report, so they cost nothing net.

The single most useful thing in this document is **Suggestion 4**. It says that the paper's fuzzy semantics is exactly the μ -cut of a max- $\otimes$ fuzzy language, which turns Section 6 from a broken comparison into a theorem, connects the paper to the whole lattice-monoid literature it currently only contrasts itself with, and makes the derivative construction a decision procedure for a well-known object rather than for a bespoke one.

The most striking is **Suggestion 6**: fuzzification can blow the minimal automaton up from $O(n)$ states to 2^n , *even for Gödel's minimum norm*. That is a genuine negative result about fuzzy matching, it is not in the literature as far as I can see, and it is exactly the kind of thing that gets a paper cited.

Notation. As in the paper: Σ finite, $\mathcal{R}$ an extensional T-similarity for a left-continuous $\otimes$, $\mathcal{R}^w$ on words, $\mathcal{R}^L$ on languages, $\mathcal{R}^E$ on restricted expressions, $L^\mu = \{w \mid \exists v \in L. \mathcal{R}^w(w, v) \geq \mu\}$, ∂_a^μ and $\mathcal{D}_a^\mu$ the fuzzy derivatives, $q_\mu : \Sigma \rightarrow \Sigma^\mu$ the quotient map. I write $\otimes^n x$ for the n -fold $\otimes$ -power and $\omega(x) = \inf_n \otimes^n x$, so that Definition 3.5's star clause is $\mathcal{R}^E(r^*, s^*) = \omega(\mathcal{R}^E(r, s))$.

Part 1 — Three theorems you get for free

These three are derivations from what is already in the paper. Each one also removes a defect from the problem report, so they are the cheapest additions on this list.

1. The recursive word-derivative equals a one-shot definition — for *every* T-norm **[VERIFIED]** **[ALSO FIXES A DEFECT]**

Statement. For $u \in \Sigma^*$ and $L \subseteq \Sigma^*$ put

$$\widehat{\partial}_u^\mu(L) = \{w \mid \exists x, v : |x| = |u|, xv \in L, \mathcal{R}^w(u, x) \otimes \mathcal{R}^w(w, v) \geq \mu\}.$$

Then $\widehat{\partial}_u^\mu(L) = \partial_u^\mu(L)$ for every u , every L , and every left-continuous $\otimes$ — no idempotence needed.

Proof. Both inclusions by induction on $|u|$. For $u = \varepsilon$ the left side is $\{w \mid \exists v \in L. \mathcal{R}^w(w, v) \geq \mu\} = L^\mu$, which is Definition 4.6's base case; for $|u| = 1$ it is Definition 4.4 verbatim.

($\supseteq$, *laxness*.) Let $M = \partial_a^{\mu''}(L)$ and $w \in \widehat{\partial}_u^{\mu'}(M)$, so there are x', v' with $|x'| = |u'|$, $x'v' \in M$ and $\mathcal{R}^w(u', x') \otimes \mathcal{R}^w(w, v') \geq \mu'$; and there are c, z with $cz \in L$ and $\mathcal{R}(a, c) \otimes \mathcal{R}^w(x'v', z) \geq \mu''$. Split

$z = z'z''$ with $|z'| = |x'|$. Take $x = cz'$ and $v = z''$; then $xv = cz \in L$ and, using the reverse triangle inequality twice and then commutativity,

$$\mathcal{R}^w(au', cz') \otimes \mathcal{R}^w(w, z'') \geq [\mathcal{R}(a, c) \otimes \mathcal{R}^w(x'v', z)] \otimes [\mathcal{R}^w(u', x') \otimes \mathcal{R}^w(w, v')] \geq \mu'' \otimes \mu'.$$

So $w \in \hat{\partial}_{au'}^{\mu' \otimes \mu''}(L) \subseteq \hat{\partial}_{au'}^\mu(L)$, the last step because $\hat{\partial}$ is antitone in the cut.

($\subseteq$.) Given a witness $x = x_1x'$, v for $w \in \hat{\partial}_{au'}^\mu(L)$, put $\alpha = \mathcal{R}(u_1, x_1)$, $\beta = \mathcal{R}^w(u', x')$, $\gamma = \mathcal{R}^w(w, v)$, so $\alpha \otimes \beta \otimes \gamma \geq \mu$. Then $x'v \in \partial_a^\alpha(L)$ — take $c = x_1$, $z = x'v$ — and (x', v) witnesses $w \in \hat{\partial}_u^{\beta \otimes \gamma}(\partial_a^\alpha(L))$. Since $\alpha \otimes (\beta \otimes \gamma) \geq \mu$, that term occurs in Definition 4.6's union. $\square$

Why it matters. This is the fix for the worst defect in the paper — Definition 4.7's sum over a continuum of cut pairs — and it is a much better fix than patching the index set, because it replaces a recursion that needs an ad-hoc finiteness argument by a definition that is manifestly a finite, single-valued construction. Three further payoffs:

- **Theorem 4.2(1) becomes a one-liner.** $\varepsilon \in \hat{\partial}_u^\mu(L)$ iff $\exists x \in L$ with $|x| = |u|$ and $\mathcal{R}^w(u, x) \otimes \mathcal{R}^w(\varepsilon, \varepsilon) \geq \mu$, i.e. iff $u \in L^\mu$. The *experimentum crucis* is then two lines rather than the paper's hardest theorem.
- The mysterious union over $\mu' \otimes \mu'' \geq \mu$ stops being a definition and becomes a *property*: the family $\{\partial_u^\mu\}$ is a strict — not merely lax — action of the quantale $([0, 1], \otimes, 1)$ on languages. See Suggestion 10.
- The same definition works on the syntactic side: $\hat{\mathcal{D}}_u^\mu(r) = \sum \mathcal{D}_x(s)$ over the finite set of (x, s) with $|x| = |u|$ and $\mathcal{R}^w(u, x) \otimes \mathcal{R}^E(s, r) \geq \mu$. That kills the continuum there too.

Cost. Two short inductions, one displayed inequality. This is half a page of paper and replaces two definitions with one.

2. $\omega(x)$ is the largest $\otimes$ -idempotent below x — which explains the star clause, and makes it computable [VERIFIED] [ALSO FIXES A DEFECT]

Statement. For a continuous $\otimes$ and $x \in [0, 1]$, $\omega(x) = \inf_n \otimes^n x = \max\{e \leq x \mid e \otimes e = e\}$. Consequently $\mathcal{R}^E(r^*, s^*)$ is the largest $\otimes$ -idempotent below $\mathcal{R}^E(r, s)$.

Proof. $\otimes^{n+1}x = (\otimes^n x) \otimes x \leq \otimes^n x$ since $\otimes \leq \min$, so the sequence decreases to some e . By continuity $\otimes^{2n}x = (\otimes^n x) \otimes (\otimes^n x) \rightarrow e \otimes e$, and it also tends to e , so e is idempotent. If $f \leq x$ is idempotent then $f = \otimes^n f \leq \otimes^n x$ for every n , so $f \leq e$. $\square$

Why it matters. Three things fall out at once.

- **The star clause becomes concrete.** For $\min$, every value is idempotent, so $\omega = \text{id}$ and $\mathcal{R}^E$ collapses to $\mathcal{G}^E$. For Goguen's product and for Łukasiewicz the idempotents are $\{0, 1\}$, so ω is the indicator of $\{1\}$: for those norms $\mathcal{R}^E(r^*, s^*)$ is 1 or 0 and nothing else. For the p -ordinal sum the idempotents are $[0, p] \cup \{1\}$, so $\omega(x) = x$ below p and p above. Stating this turns an opaque infimum into a table.
- **Decidability.** The problem report flags that $\mathcal{R}^E(s, r) \geq \mu$ need not be decidable because of the infimum. With ω given as “largest idempotent below”, it is: the idempotents of a continuous T-norm are the endpoints of its ordinal-sum intervals, and for the four norms the paper names they are finite or a single interval.
- **It explains the paper's recurring hypothesis.** If μ is idempotent then $\mu \leq \omega(x) \iff \mu \leq x$, so the μ -cut cannot tell r^* from r — which is exactly why the star clause stops mattering under $\mu \otimes \mu = \mu$, and why Remark 4.2 computes different derivatives for $\min$ and for the product norm.

Cost. A five-line lemma plus a four-row table. Worth putting in Section 3 immediately after Definition 3.5.

3. Under $\mu \otimes \mu = \mu$, the μ -cut of $\mathcal{R}^E$ is exactly “same skeleton, positionwise μ -similar” [VERIFIED] [ALSO FIXES A DEFECT]

Statement. Let $\mu \otimes \mu = \mu$ and let r, s be restricted. Then

$$\begin{aligned} \mathcal{R}^E(r, s) \geq \mu &\iff r = r_0[a_1, \dots, a_n] \text{ and } s = r_0[b_1, \dots, b_n] \text{ with } \mathcal{R}(a_i, b_i) \geq \mu \text{ for every } i \\ &\iff q_\mu^*(r) = q_\mu^*(s). \end{aligned}$$

Proof. By induction on the common skeleton. Atoms are immediate. For $+$ the clause is $\min$, so the value is $\geq \mu$ iff both are. For $\cdot$ the clause is $\otimes$: forwards because $\otimes \leq \min$, backwards because $\mu \otimes \mu = \mu$. For $*$ use Suggestion 2: μ is idempotent, so $\mu \leq \omega(x) \iff \mu \leq x$. The second equivalence is the observation that q_μ^* preserves the tree shape and replaces letters by classes. $\square$

Why it matters. This is the lemma Section 4.3 is missing, and it does a surprising amount of work:

- **Definition 4.5 becomes explicit and obviously finite.** Under the hypothesis,

$$\mathcal{D}_a^\mu(r) = \sum_{b \in [a]_\mu} \sum_{q_\mu^*(s) = q_\mu^*(r)} \mathcal{D}_b(s),$$

a sum over $|[a]_\mu| \cdot \prod_i |[a_i]_\mu|$ terms. Conjecture 3.1 and Lemma 3.3 are then not needed at all in the idempotent case, and the index set is decidable by inspection rather than by evaluating an infimum.

- It is the missing step in Proposition 4.5 and in the quotient theorem, both of which silently use “ $\mathcal{R}^E(s, r) \geq \mu$ implies $[s]_\mu = [r]_\mu$ ”.
- It is the right definition to give Rocq: a structural, decidable, boolean test, with no reals in it at all.

Cost. One induction. It should be stated as a lemma in Section 3 and used four times in Section 4.3.

Part 2 — Four results that would make the paper stand out

4. The paper’s fuzzy semantics is the μ -cut of a $\max\text{-}\otimes$ fuzzy language [VERIFIED]

Statement. For a restricted r define $\|r\| : \Sigma^* \rightarrow [0, 1]$ by $\|r\|(w) = \sup\{\mathcal{R}^w(w, v) \mid v \in \mathcal{L}(r)\}$. Then $\|\cdot\|$ satisfies

$$\begin{aligned} \|\emptyset\| &= 0, \quad \|\varepsilon\| = \mathbf{1}_{\{\varepsilon\}}, \quad \|a\|(w) = \begin{cases} \mathcal{R}(w, a) & |w| = 1 \\ 0 & \text{else,} \end{cases} \\ \|r + s\| &= \max\{\|r\|, \|s\|\}, \quad \|r \cdot s\|(w) = \sup_{w=uv} \|r\|(u) \otimes \|s\|(v), \\ \|r^*\|(w) &= \sup_{k \geq 0} \sup_{w=w_1 \dots w_k} \bigotimes_i \|r\|(w_i), \end{aligned}$$

and $L^\mu(r) = \{w \mid \|r\|(w) \geq \mu\}$. That is: $\|\cdot\|$ is precisely the standard $\max\text{-}\otimes$ fuzzy-language semantics of r , with each atom a interpreted as the fuzzy singleton $\mathcal{R}(\cdot, a)$.

Proof. The $+$, $\emptyset$, ε and atom clauses are immediate. For $\cdot$: every $z \in \mathcal{L}(r)\mathcal{L}(s)$ factors as $z_1 z_2$, and $\mathcal{R}^w$ is 0 unless lengths agree, so $\mathcal{R}^w(w, z) = \mathcal{R}^w(w, z_1) \otimes \mathcal{R}^w(w, z_2)$ for the split of w at $|z_1|$;

taking suprema and using left-continuity of $\otimes$ to pull $\otimes$ through sup gives the clause. Star is the same computation over all k , with the empty product equal to 1. The cut identity holds because Σ is finite and $\mathcal{R}^w$ vanishes off equal lengths, so the supremum is over a finite set and is attained. $\square$

Why it matters. This is the observation I would build the revised paper around.

- **It repairs Section 6 and turns it into a theorem.** Section 6 currently tries to relate the paper to the scalar approach of [LP06, SC12, GJT17] by adjoining an uncountable alphabet $\{a^\lambda\}$ and a relation that is not even reflexive. None of that is needed: the similarity-based construction *is* the lattice-monoid construction, instantiated at the fuzzy sets $\mathcal{R}(\cdot, a)$. Say that, and the comparison is one paragraph and correct.
- **It makes the main theorem a statement about a known object.** Theorem 4.2(1) reads $\|r\|(u) \geq \mu \iff \varepsilon \in \mathcal{L}(\mathcal{D}_u^\mu(r))$: *Brzozowski derivatives decide threshold membership for max- $\otimes$ weighted regular languages*. That is a sentence an automata-theory audience will recognise and care about, and it is strictly more informative than “ $u \in L^\mu(r)$ ”.
- **It explains the two inclusions of Theorem 4.2(3)–(4).** $\mathcal{D}_u^\mu(r)$ is a crisp under-approximation of the *residual* $\|r\|(u \cdot)$, and $\mathcal{L}(\mathcal{D}_u^\mu(r)) \subseteq \partial_u^\mu(\mathcal{L}r) \subseteq L^\mu(\mathcal{D}_u^\mu(r))$ says the approximation is tight to within one further μ -step. Stated that way, the pair of inclusions looks like a feature rather than a weakened equality.
- **It is what makes the theory genuinely fuzzy.** As written, the paper’s objects are all crisp: $L^\mu(r)$ is an ordinary language. $\|r\|$ is an actual fuzzy language, and it is already implicit in Definition 3.3 — $\mathcal{R}^L(\mathcal{L}(r), \mathcal{L}(s)) = \min\{\inf_{w \in \mathcal{L}(r)} \|s\|(w), \inf_{w \in \mathcal{L}(s)} \|r\|(w)\}$.

Cost. One definition, one lemma with the six clauses, and a rewritten Section 6. Half the work is already done in Definitions 3.2 and 3.3.

5. The idempotence hypothesis is not a convenience — it is a characterization [VERIFIED]

Statement. Let $\otimes$ be a left-continuous T-norm and $0 < \mu \leq 1$. The following are equivalent.

- (i) $\mu \otimes \mu = \mu$;
- (ii) for every Σ , every extensional T-similarity $\mathcal{R}$ on Σ and every $L \subseteq \Sigma^*$, $(L^\mu)^\mu = L^\mu$ — that is, $(\cdot)^\mu$ is a closure operator;
- (iii) for every such $\Sigma, \mathcal{R}$, the relation $\sim_\mu^{\Sigma^*}$ is transitive;
- (iv) for every such $\Sigma, \mathcal{R}$ whose μ -cut is an equivalence on Σ — so that the quotient exists — and every L and every a , $\partial_a^\mu(L) = (q_\mu^*)^{-1}(\partial_{[a]_\mu}(q_\mu^*(L)))$.

Proof idea. (i) $\Rightarrow$ (ii),(iii),(iv) are the paper’s own arguments with the hypothesis made explicit; (iv) needs $\otimes^n \mu = \mu$, which follows from (i) by induction. For the converses, suppose $\mu \otimes \mu < \mu$; then $\mu < 1$, since $1 \otimes 1 = 1$. A single two-letter witness refutes all three. Take $\Sigma = \{a, b\}$ with $\mathcal{R}(a, a) = \mathcal{R}(b, b) = 1$ and $\mathcal{R}(a, b) = \mu$. This is an extensional T-similarity, and its μ -cut is all of $\Sigma \times \Sigma$, hence an equivalence, so Σ^μ is the one-letter alphabet and (iv) is well posed. Then:

- *(ii) fails.* For $L = \{aa\}$ we get $L^\mu = \{aa, ab, ba\}$, since $\mathcal{R}^w(aa, bb) = \mu \otimes \mu < \mu$; but $bb \in (L^\mu)^\mu$, because $\mathcal{R}^w(ab, bb) = \mu \otimes 1 = \mu$.
- *(iii) fails.* $aa \sim_\mu ab \sim_\mu bb$ while $aa \not\sim_\mu bb$.
- *(iv) fails.* For $L = \{aaa\}$, the right-hand side is all of Σ^2 , while $\partial_a^\mu(L) = \{aa, ab, ba\}$ — bb is again excluded by $\mu \otimes \mu < \mu$.

$\square$

Why it matters. The problem report lists eight separate places where the paper needs $\mu \otimes \mu = \mu$ and does not say so, and the natural reading is that the fuzzy theory is fragile. This theorem says the opposite: the hypothesis is *exactly* the condition under which a fuzzy Brzowski theory can exist at all, on the nose, and the two-letter witness shows nothing weaker will do. Sharpness results of this shape are what referees ask for, and here it is cheap: the witness is one line of data and three of arithmetic.

It also gives Section 4.3 a reason to exist other than “the special case we can handle”: under (i)–(iv) the μ -neighbourhood is a Kuratowski closure operator, so Σ^* carries a topology whose closed sets are the μ -closed languages, and the whole fuzzy theory is the crisp theory of that topology. See Suggestion 8.

Cost. One theorem and one two-letter counterexample. It replaces four verbatim repetitions of a compound hypothesis.

6. Fuzzification can cost an exponential number of states — even for Gödel’s norm [VERIFIED]

Statement. Assume $\mu \otimes \mu = \mu$ and let r have a minimal DFA with n states.

- (a) The minimal DFA of $L^\mu(r)$ has at most 2^n states.
- (b) This is tight. There is a family with $\otimes = \min$, $|\Sigma| = 3$, minimal DFA of size $n + 2$, and minimal DFA of $L^\mu(r)$ of size exactly 2^n .
- (c) If $\mathcal{R}(a, b) \geq \mu$ for *all* a, b , then $L^\mu(r) = \{w \mid |w| \in \text{len}(\mathcal{L}(r))\}$ depends only on word length, and the state count is the preperiod plus period of the length spectrum of $\mathcal{L}(r)$ — which for an n -state DFA can be as large as Landau’s function $e^{\Theta(\sqrt{n \log n})}$.

Proof. (a) By Suggestion 3, $\mathcal{R}^w(w, v) \geq \mu$ iff $q_\mu^*(w) = q_\mu^*(v)$, so $L^\mu(r) = (q_\mu^*)^{-1}(q_\mu^*(\mathcal{L}(r)))$. Relabelling the transitions of the DFA of $\mathcal{L}(r)$ along q_μ gives an n -state NFA for $q_\mu^*(\mathcal{L}(r))$, so that language has a minimal DFA of at most 2^n states. And since q_μ^* is surjective, the Myhill–Nerode classes of $(q_\mu^*)^{-1}(M)$ are exactly the q_μ^* -preimages of those of M , so the two minimal DFAs have the *same* number of states.

(b) Let $\Sigma = \{a_0, a_1, b\}$ with $\otimes = \min$, $\mathcal{R}(a_0, a_1) = \mu$ and all other off-diagonal values 0; then $\Sigma^\mu = \{[a], [b]\}$ is binary. Let D have states $0, 1, \dots, n$ plus a sink, with $0 \xrightarrow{a_0} 0$, $0 \xrightarrow{b} 0$, $0 \xrightarrow{a_1} 1$, and $i \xrightarrow{\cdot} i + 1$ for $1 \leq i < n$ on every letter, n accepting and absorbing into the sink. D is deterministic with $n + 2$ states, and $q_\mu^*(\mathcal{L}(D)) = ([a] + [b])^* [a] ([a] + [b])^{n-1}$: the letter a_1 marks which occurrence of $[a]$ is the one n positions from the end. That language is the standard “ n th-symbol-from-the-end” language, whose minimal DFA has 2^n states. By (a) so does $L^\mu(D)$.

(c) With every pair μ -similar, Σ^μ is a single letter, so $q_\mu^*(w)$ records only $|w|$; the rest is the standard theory of unary regular languages, and the bound is realised by taking $\mathcal{L}(r) = \bigcup_i a_i^{p_i \mathbb{N}}$ for distinct primes p_i with $\sum_i p_i = n$, whose length spectrum has period $\prod_i p_i$. $\square$

Why it matters. This is the result I would lead the abstract with, because it is a real negative result and it is counter-intuitive: matching *up to a similarity* sounds like it should be easier than exact matching, and it is easier nondeterministically, but deterministically it can cost exponentially more. It also gives the paper’s finiteness theorems (Theorem 4.2(4),(5)) a quantitative counterpart, which they currently lack entirely — “finite” is a much weaker statement than “finite, and here is the tight bound”.

Note also where the cost sits: by Suggestion 11 the fuzzy *nondeterministic* automaton stays linear. So the derivative construction localises the blow-up precisely at determinisation, which is a clean story and is exactly what the derivative-of-a-sum shape of $\mathcal{D}_a^\mu$ is telling you.

Cost. One page, and part (a) is three lines once Suggestion 3 is in place.

7. μ -equivalence of two DFAs is Pspace-complete, while ordinary equivalence is in P [PROOF SKETCH]

Statement. Fix $\otimes = \min$ and $0 < \mu < 1$. Given two DFAs A, B over Σ and an extensional T-similarity $\mathcal{R}$, deciding $L^\mu(A) = L^\mu(B)$ is PSPACE-complete. Deciding $\mathcal{L}(A) = \mathcal{L}(B)$ is in P. Fuzzy *membership*, by contrast, stays linear in $|u|$.

Proof idea. Membership in PSPACE: by Suggestion 6(a) the question is equivalence of the two relabelled NFAs, which is in PSPACE. Hardness: NFA equivalence is PSPACE-complete, and every n -state NFA N over Γ is the q_μ -image of a DFA — take $\Sigma = \Gamma \times T$ where T names transitions, let the DFA follow the named transition, and let $\mathcal{R}$ have μ -classes $\{(g, t) \mid t \in T\}$ for each $g \in \Gamma$, which is realisable by Suggestion 6’s construction. Two NFAs can be encoded over a common Σ , so the reduction is uniform. $\square$

Why it matters. A complexity separation is the most compact way to say that fuzzification is not a cosmetic change. And the contrast within the same theorem is the memorable part: membership stays cheap, equivalence jumps two complexity classes. It also tells a reader why the paper’s derivative construction is the right tool — it gives the cheap operation directly, and it is honest about the expensive one.

Cost. Half a page, given Suggestion 6. The only thing to check carefully is that the reduction can be made to use one fixed $(\mathcal{R}, \mu)$ for both machines.

Part 3 — Structure worth adding

8. The μ -closed languages form a Kleene algebra with complement, isomorphic to $\text{Reg}((\Sigma^\mu)^*)$ [VERIFIED]

Statement. Assume $\mu \otimes \mu = \mu$ and call L μ -closed when $L = L^\mu$. Then L is μ -closed iff $L = (q_\mu^*)^{-1}(M)$ for some $M \subseteq (\Sigma^\mu)^*$, and $(q_\mu^*)^{-1}$ is an isomorphism from the languages over Σ^μ onto the μ -closed languages over Σ that preserves $\cup$, $\cap$, complement, concatenation, star and the derivative:

$$(q_\mu^*)^{-1}(M) \cdot (q_\mu^*)^{-1}(N) = (q_\mu^*)^{-1}(M \cdot N), \quad \partial_a((q_\mu^*)^{-1}(M)) = (q_\mu^*)^{-1}(\partial_{[a]_\mu}(M)).$$

It restricts to an isomorphism on the regular languages, and $L^\mu(r)$ is always μ -closed.

Proof. Inverse image commutes with all Boolean operations. For concatenation, $\subseteq$ is immediate; for $\supseteq$, if $q_\mu^*(w) \in M \cdot N$ then, because q_μ^* is length-preserving, the factorisation of $q_\mu^*(w)$ lifts to a factorisation of w at the same position. Star is the same argument iterated. The derivative clause is the quotient theorem (Proposition 4.4) restated. $\square$

Why it matters. The problem report’s third architectural option was “quotient-first”: lead with the quotient theorem and derive everything from it. This is the theorem that licenses it. It says the fuzzy theory is not merely *reducible* to the crisp theory over Σ^μ ; it *is* that theory, as an isomorphism of algebras with complement — which is stronger than Proposition 4.4 and much easier to remember. Reorganising Section 4.3 around it would let three of the paper’s propositions become one-line corollaries.

Cost. One lemma. It is the single best return on rewriting effort in the paper.

9. Skeleton normal form for $\mathcal{R}^E$ [VERIFIED]

Statement. For every restricted r_0 with n symbol occurrences there is a term P_{r_0} in the operations $\min$, $\otimes$ and ω , read off the syntax tree of r_0 , such that

$$\mathcal{R}^E(r_0[a_1, \dots, a_n], r_0[b_1, \dots, b_n]) = P_{r_0}(\mathcal{R}(a_1, b_1), \dots, \mathcal{R}(a_n, b_n)),$$

and $\mathcal{R}^E(r, s) = 0$ whenever r and s have different skeletons. Moreover $P_{r_0}(\vec{x}) \leq \min_i x_i$.

Proof. Immediate induction on r_0 : $+$ contributes $\min$, $\cdot$ contributes $\otimes$, $*$ contributes ω . The bound holds because each of $\min$, $\otimes$ and ω is $\leq$ each of its arguments. $\square$

Why it matters. Small, but it does four jobs. It *is* Conjecture 3.1, proved. It makes Lemma 3.3 a counting statement ($|\Sigma|^n$ candidates). It gives the “ $\mathcal{R}^E$ is at most the minimum of its leaf values” fact that the quotient results use silently. And it is the right thing to define in Rocq, because it separates the combinatorics (skeletons, computable) from the value algebra (the T-norm, real-valued) — which is exactly the seam where the current development gets stuck.

Cost. Ten lines. Do it when fixing Conjecture 3.1.

10. The fuzzy derivative is a quantale action, and Suggestion 1 says it is a strict one [VERIFIED]

Statement. The family $\{\partial_u^\mu\}_{u,\mu}$ satisfies $\partial_v^{\mu'} \circ \partial_a^{\mu''} \subseteq \partial_{av}^{\mu' \otimes \mu''}$, is antitone in the cut, and has $\partial_\varepsilon^1 = \text{id}$: it is a lax action of the quantale $([0, 1], \otimes, 1)$ on $\mathcal{P}(\Sigma^*)$. By Suggestion 1 the induced μ -graded composite is in fact *equal* to ∂_{av}^μ , so the action is strict.

Proof idea. The inclusion is the $(\supseteq)$ half of Suggestion 1’s proof; antitonicity is immediate; strictness is Suggestion 1.

Why it matters. This is presentational, but it is the presentation that makes Section 4 readable. At the moment the reader meets a union over $\{(\mu', \mu'') \mid \mu' \otimes \mu'' \geq \mu\}$ with no explanation of where it comes from, and it looks like a hack. Saying “the fuzzy derivative is the $[0, 1]$ -graded Brzozowski derivative, and the union is the composition law of the grading” makes it inevitable instead of arbitrary — and it connects the paper to the graded-monad and quantale-enriched literature at zero mathematical cost. It also pairs well with Suggestion 5: the grading is degenerate exactly on the idempotent cuts.

Cost. A remark and a two-line lemma.

11. Fuzzy partial derivatives: the nondeterministic automaton stays linear [PROOF SKETCH]

Statement. Define the fuzzy partial-derivative set $\Delta_a^\mu(r)$ as the *set* of Antimirov partial derivatives $\partial_b(s)$ over the index set of Definition 4.5, rather than their sum. Then, under $\mu \otimes \mu = \mu$, the reachable set of partial derivatives of r has at most $m + 1$ elements, where m is the number of symbol occurrences in r , and the resulting NFA over Σ accepts exactly $L^\mu(r)$.

Proof idea. By Suggestion 3 the index set is $\{(b, s) \mid [b]_\mu = [a]_\mu, q_\mu^*(s) = q_\mu^*(r)\}$, and all those s have the same skeleton as r , so their partial derivatives are the partial derivatives of $q_\mu^*(r)$ read back along q_μ . Antimirov’s bound for $q_\mu^*(r)$ then applies, and q_μ^* does not change the number of symbol occurrences.

Why it matters. Two reasons to want this. First, it removes the problem report’s objection that the sum in Definition 4.5 is only well defined up to associativity, commutativity and idempotence of $+$: a *set* of expressions has no such ambiguity, which makes the definition honest on paper and trivial in Rocq. Second, together with Suggestion 6 it gives the paper a complete complexity

picture — linear nondeterministic, exponential deterministic, and the exponential is exactly the determinisation — which is a much better story than either result alone.

Cost. A definition and a lemma, plus a short proof that the NFA is correct. The Antimirov bound is standard.

12. Say what $\mathcal{R}^L$ is: a fuzzy Hausdorff distance [VERIFIED] [ALSO FIXES A DEFECT]

Statement. $\mathcal{R}^L$ is the Hausdorff-style lifting of $\mathcal{R}^w$: with $\|r\|$ as in Suggestion 4, $\mathcal{R}^L(L_1, L_2) = \min\{\inf_{w \in L_1} \|L_2\|(w), \inf_{w \in L_2} \|L_1\|(w)\}$, where $\|L\|(w) = \sup_{v \in L} \mathcal{R}^w(w, v)$. Under the conventions $\inf \emptyset = 1$, $\sup \emptyset = 0$ it is an extensional T-similarity on $\mathcal{P}(\Sigma^*)$ whenever Σ is finite, and left-continuity of $\otimes$ is used exactly once, in the reverse triangle inequality.

Proof. The reverse triangle inequality is the standard Hausdorff argument, with $\otimes$ pulled through the suprema by left-continuity; extensionality needs the suprema to be attained, which holds because $\mathcal{R}^w$ vanishes off equal lengths and Σ is finite. $\square$

Why it matters. Purely expository, but cheap and worth it. Naming the construction tells the reader in one word what Definition 3.3 is, explains why the two nested inf sup terms are there, and explains why left-continuity is assumed — which the paper currently never says. It also gives the honest statement of where finiteness of Σ is genuinely used, as opposed to the four places where it is merely repeated.

Cost. Two sentences and a footnote.

Part 4 — Two open problems worth stating

A paper that states its open problems precisely is easier to cite than one that does not. Both of these are natural, both are within reach of the paper’s own machinery, and neither is embarrassing to leave open.

13. Which $\max\text{-}\otimes$ fuzzy regular languages are similarity-generated? [OPEN]

Statement. By Suggestion 4 every restricted r and similarity $\mathcal{R}$ give a fuzzy regular language $\|r\|$ in the $\max\text{-}\otimes$ sense. Characterise the image: which fuzzy regular languages $\varphi : \Sigma^* \rightarrow [0, 1]$ are of the form $\|r\|$ for some Σ , some extensional T-similarity $\mathcal{R}$ and some restricted r ?

Why it matters. This is the sharp form of the comparison Section 6 is groping towards, and it is the question a reader from the lattice-monoid tradition will immediately ask. Two easy necessary conditions come free: φ is 1 exactly on $\mathcal{L}(r)$ (by extensionality), and φ is constant on each $\sim_\mu$ class for every μ — so the candidates are the “similarity-invariant” fuzzy regular languages. Whether that is sufficient is exactly the open part. Even a partial answer for $\otimes = \min$ would be a result.

14. Does regularity of $\partial_a^\mu(L)$ survive a non-idempotent cut? [OPEN]

Statement. For $\mu \otimes \mu < \mu$ and L regular, is $\partial_a^\mu(L)$ regular?

Why it matters. The problem report explains why this is not routine: the membership test is a threshold on a $\otimes$ -product along the word, so $\partial_a^\mu(L)$ is a cut-point language of a $\max\text{-}\otimes$ weighted automaton, and cut-point languages of weighted automata are not regular in general. My own reading is that it *is* regular for every continuous $\otimes$, by a bounded-budget argument: factors that are $\otimes$ -idempotent are free and reduce to a letterwise condition, and within each Archimedean piece of the ordinal-sum decomposition only boundedly many non-idempotent factors above μ can occur before the product drops below μ , so a finite automaton can carry the remaining budget. For a left-continuous but discontinuous $\otimes$ — the nilpotent minimum, say — I do not know. Stating it as a proposition for continuous norms and a question in general is both honest and useful, and

it is the one place in the paper where the general- $\otimes$ setting has a genuinely open mathematical question attached to it, which is an asset rather than a gap.

Part 5 — What I would not chase

- **Minimality of the fuzzy derivative automaton.** It is true and it is a corollary of Suggestion 8 (the minimal DFA of $L^\mu(r)$ is the pullback of the minimal DFA of $q_\mu^*(\mathcal{L}(r))$), so there is nothing to prove beyond stating it. The README.md currently lists a “DFA minimality theorem” that does not exist in the paper; this is what it should be.
- **A complete axiomatisation of μ -equivalence.** By Redko there is none in the crisp case, and Suggestion 7 says the μ -case is strictly harder, so this is a dead end.
- **The $\#$ substitution operator of Appendix B.** It deliberately leaves the regular languages and nothing else in the paper uses it. Either drop it or make it a separate note.
- **Extending $\mathcal{R}^E$ to $\&$ and $\neg$.** The paper’s own two counterexamples show why, and the $\neg$ one is norm-independent. State it as a proposition — “no extension of Definition 3.5 to $\neg$ satisfies Theorem 3.1(1)” — and stop there.
- **Fuzzy transition functions.** The remark at the very end of the dead block suggests reading $\delta(q, [a]) = q'$ as graded membership. Suggestion 4 already gives the paper a genuinely graded object, $\|r\|$, and it is a better one; a graded δ on top of a crisp quotient automaton would be decoration.

Part 6 — What this does to the paper

If Suggestions 1–4 and 8 go in, most of the current fuzzy section becomes corollaries, which is a good sign rather than a bad one:

Current result	Becomes
Definitions 4.6 and 4.7 (recursive, one over a continuum)	one non-recursive definition, plus Suggestion 1 as a theorem
Theorem 4.2(1), the <i>experimentum crucis</i>	two lines from Suggestion 1
Theorem 4.2(3)–(4), the two inclusions	a statement about how well a crisp derivative approximates a residual (Suggestion 4)
Conjecture 3.1 and Lemma 3.3	Suggestion 9
Proposition 4.1 and Proposition 4.3	corollaries of Suggestion 3
Proposition 4.4, the quotient theorem	one clause of the isomorphism in Suggestion 8
The four repetitions of “gödelian, or $\mu \otimes \mu = \mu$ ”	one named hypothesis, justified by Suggestion 5
Section 6	one paragraph, correct, plus the open problem of Suggestion 12

And the paper gains three things it does not currently have: a quantitative theorem (Suggestion 6), a complexity theorem (Suggestion 7), and a sharpness theorem (Suggestion 5). Those are the three kinds of result that make a derivative paper stand out from the existing fuzzy-automata literature, which is almost entirely definitional.

Everything tagged **[VERIFIED]** I proved from the paper’s own definitions while writing this, including the counterexamples in Suggestions 5 and 6; the arithmetic in Suggestion 6(b) was checked against the standard 2^n lower bound for the n th-symbol-from-the-end language. Everything tagged **[PROOF SKETCH]** rests on a standard fact I did not re-derive — Antimirov’s bound in Suggestion 11, and PSPACE-completeness of NFA equivalence in Suggestion 7. Suggestions 12 and 13 are questions; the reading I give of 13 is my own and is not proved.